Fiscal Year:	FY 2021	**Task Last Updated:**	FY 07/22/2021
PI Name:	Mias, George Ph.D.		
Project Title:	Integrative Personalized Omics Profiling Next Steps: Detection and Classification of Deviations from Wellness		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	Yes
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	george.mias@gmail.com	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	517-353-0855
Organization Name:	Michigan State University		
PI Address 1:	Biochemistry and Molecular Biology		
PI Address 2:	775 Woodlot Dr, Rm 1319		
PI Web Page:			
City:	East Lansing	**State:**	MI
Zip Code:	48824	**Congressional District:**	8
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	12/31/2020
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Piermarocchi, Carlo Ph.D. (Michigan State University)		
Grant/Contract No.:	NNX16AO69A-T0412		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

Precision medicine is an emergent interdisciplinary field that utilizes state-of-the art technologies to detect changes in individual wellness, diagnose, treat, and possibly prevent diseases. Previous studies, by us and others, have focused on establishing the feasibility of personalized precision medicine, by measuring multiple time signals in individuals (generalized omics). These signals typically include medically relevant measures, such as clinical tests, blood panels, gene expression, and data from personal health devices. However, algorithmic development and sophisticated methods to analyze, classify, and biologically annotate such omics are still underdeveloped. In our proposed investigation, we will focus on the essential next steps to implement such integrative Personal Omics Profiling (iPOP) on individual astronauts, and detect and classify adverse medical events through the analysis of generalized omics signals.
Objectives: Our objectives are to: (i) Generate new computational methodology to establish the baseline healthy characteristics in individual personnel, and detect baseline deviations that predict the onset of any adverse medical event. (ii) Classify departures from individualized wellness according to collective biological/physiological signal characteristics and behavior. We will annotate the classes based on medical relevance, for instance by associating groups of signals to a specific disease. Our methodology will compare medical baseline deviations both within and across individuals.

Methods: To achieve our objectives we plan to develop sophisticated computational methods in statistical time series analysis and dynamic network theory to characterize and categorize different biological/physiological signals from individuals. We will develop and implement new algorithms that establish an individual's health baseline through the integration of thousands of signals, and that can detect deviations from a person's own baseline. The time series will be analyzed in terms of their underlying frequencies and limitations due to missing data or uneven sampling will be addressed. We will create a geometric representation of the signals, including networks that represent biological interactions between the omics that will allow us to both detect anomalies, but also infer the medical relevance of abnormal medical events.

Significance to solicitation's objectives: Our proposed investigation is completely aligned with Topic 1 of the solicitation, to develop Predictive algorithms of health, behavior, and medical events. The frameworks and methods of this investigation are designed to be directly applied to the monitoring of spaceflight personnel. Deliverables of this project include: (i) novel computational methodology, (ii) open source software packages (including a standalone analysis tool), and (iii) a report assessing the predictive utility of different data types and tissue samples that may be used in flight (including saliva).

Significance to NASA's interests and programs: Personalized monitoring of astronauts is already being performed, and through implementing our novel methods we will be able to improve NASA's ability to detect baseline abnormalities in flight, intervene with corrective measures, and maintain individualized wellness for personnel, which is mission critical.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

We have continued to develop new algorithms that can identify changes in health, and potential medical events. All our methods are extensible and can be used to incorporate any dynamic data that can be measured during spaceflight - including different sampling rates (this includes molecular as well as device data that can be quantified). Furthermore, our approach can be implemented on Earth as well using personnel medical monitors to detect medical events.
The frameworks and methods of this investigation were primarily designed to be applied to the monitoring of individual spaceflight personnel. We have extended our trend detection algorithms (implemented in the open source Python package PyIOmica, and updated in Year 2 with new functionality and python compatibility) that address multiple challenges faced in analyzing signal data from human subjects and in-mission personnel. Specifically our trend detection algorithms and community detection methods:

(i) integrate thousands of time series per person,

(ii) apply to multiple signals of varying origin (e.g., transcripts versus metabolites versus radiation measurements),

(iii) address different time frames over which measurements may be taken (e.g., seconds versus hourly or daily measurements),

(iv) use many or few time points in the analysis,

(v) address missing data,

(vi) address uneven sampling of signals (typically the case for in-flight or in-the-field measurements),

(vii) integrate multiple layers of information (from different measurements, using different monitors),

(viii) provide utilities for visualization and facilitating the interpretation of data by non-specialists (including pathway annotations for gene-based or metabolite-based measurements),

(ix) provide visibility graphs methodology to detect and visualize temporal communities corresponding to changing physiological states in time series signals.

The methods can be used to analyze individualized changes (per subject) compared to their own baseline, and also provide data that can be used to compare between-subject changes to identify common differential changes corresponding to potential medical events that show common responses across subjects.

Deliverables of this project include: (i) novel computational methodology, (ii) open source software packages (including a standalone analysis tool), and (iii) assessing the predictive utility of different data types and tissue samples that may be used in flight (including saliva).

We have implemented the analysis to data for validations including:

(i) the evaluation of saliva multi-omics for non-invasive diagnostics and evaluation of predictive information per omics-data type analyzed in an individual subject to immune activation via vaccination.

(ii) the evaluation of individual subjects subject to immune activation (using retrospective data from prediabetic cohort analyisis).

(iii) the evaluation of retrospective astronaut data, from long-duration missions to assess the effects of spaceflight, and detect events that may be associated to medical in-flight events.

Significance to NASA's interests and programs: Personalized monitoring of astronauts is already being performed, and through implementing our novel methods we will be able to improve NASA's ability to detect baseline abnormalities in flight, intervene early with corrective measures, and maintain individualized wellness for personnel, which is mission critical. Furthermore, saliva omics offers non-invasive diagnostics, which can potentially be extended to spaceflight monitoring in long-duration space missions.

Task Progress:

Precision medicine is an emergent interdisciplinary field that utilizes state-of-the art technologies to detect changes in individual wellness, diagnose, treat, and possibly prevent diseases. Previous studies, by us and others, have focused on establishing the feasibility of personalized precision medicine, by measuring multiple time signals in individuals (generalized omics). These signals typically include medically relevant measures, such as clinical tests, blood panels, gene expression, and data from personal health devices. However, algorithmic development and sophisticated methods to analyze, classify, and biologically annotate such omics are still underdeveloped. In our proposed investigation, we will focus on the essential next steps to implement such integrative Personal Omics Profiling (iPOP) on individual astronauts, and detect and classify adverse medical events through the analysis of generalized omics signals.
Objectives: Our objectives are to: (i) Generate new computational methodology to establish the baseline healthy characteristics in individual personnel, and detect baseline deviations that predict the onset of any adverse medical event. (ii) Classify departures from individualized wellness according to collective biological/physiological signal characteristics and behavior. We will annotate the classes based on medical relevance, for instance by associating groups of signals to a specific disease. Our methodology will compare medical baseline deviations both within and across individuals.

Methods: To achieve our objectives we plan to develop sophisticated computational methods in statistical time series analysis and dynamic network theory to characterize and categorize different biological/physiological signals from individuals. We will develop and implement new algorithms that establish an individual's health baseline through the integration of thousands of signals, and that can detect deviations from a person's own baseline. The time series will be analyzed in terms of their underlying frequencies and limitations due to missing data or uneven sampling will be addressed. We will create a geometric representation of the signals, including networks that represent biological interactions between the omics that will allow us to both detect anomalies, but also infer the medical relevance of abnormal medical events.

Significance to solicitation's objectives: Our proposed investigation is completely aligned with Topic 1 of the solicitation, to develop Predictive algorithms of health, behavior, and medical events. The frameworks and methods of this investigation are designed to be directly applied to the monitoring of spaceflight personnel.

Deliverables of this project include: (i) novel computational methodology, (ii) open source software packages (including a standalone analysis tool), and (iii) a report assessing the predictive utility of different data types and tissue samples that may be used in flight (including saliva).

Significance to NASA's interests and programs: Personalized monitoring of astronauts is already being performed, and through implementing our novel methods we will be able to improve NASA's ability to detect baseline abnormalities in flight, intervene with corrective measures, and maintain individualized wellness for personnel, which is mission critical.

Key Findings (Year 1):

Objective (i): We have developed an algorithm for adverse event detection that can characterize baseline deviations from any time series signal. We have implemented the algorithm in the published Python package PyIOmica. The algorithm can integrate thousands of time series with different normalizations and time frames, and can address issues of missing data and uneven sampling. For measurements (time-series) with gene-annotated information, and/or metabolite data, pertinent pathways that are affected can be identified to assess biological and medical significance. Furthermore the algorithm can be used to incorporate any time-signal from different medical monitors used in-flight. The algorithm will act as a backend base for the deliverable AstroIO application that will integrate multimodal information from various health monitors in Year 2. We have implemented the methodology leading to Aim 1 to classify multi-omics measurements in saliva from a single individual as a proof of principle for saliva utility in non-invasive diagnostic monitoring of physiological state and adverse events (hourly and daily monitoring using saliva RNA/protein and extracellular vesicles).

Key Findings (Year 2):

Objective (ii). We have completed new algorithms that can classify departures from individualized wellness according to collective biological/physiological signal characteristics and behavior. Our methodology compares medical baseline deviations both within and across individuals. We compared individualized wellness departures in a) omics data that we have generated from saliva b) omics data obtained from prediabetics and diabetics, and c) data obtained from Lifetime Surveillance of Astronaut Health (LSAH) from two previous studies on astronauts (data obtained mid-Year 2). We annotated and detected various medical events using our methodology. Multiple layers of time series signals from different measurements were integrated, and we built new algorithms to detect changepoints between changing physiological states in individuals and to compare these results across individuals. The biological/medical significance of the data were assessed through pathway annotations of networks. The algorithms are being integrated in the PyIOmica package, and will be incorporated into the open source desktop application AstroIO (javascript). Retrospective astronaut data were used to apply the algorithms and for categorizing already observed medical events. Finally, we have published our findings on the utility of saliva for non-invasive diagnostics.

Bibliography Type: Description: (Last Updated: 10/30/2023)

Articles in Peer-reviewed Journals

Domanskyi S, Piermarocchi C, Mias GI. "PyIOmica: longitudinal omics analysis and trend identification." Bioinformatics. 2020 Apr;36(7):2306-7. https://doi.org/10.1093/bioinformatics/btz896 ; PMID: 31778155; PMCID: PMC7141865 , Apr-2020

Articles in Peer-reviewed Journals	Mias GI, Singh VV, Rogers LRK, Xue S, Zheng M, Domanskyi S, Kanada M, Piermarocchi C, He J. "Longitudinal saliva omics responses to immune perturbation: a case study." Sci Rep. 2021 Jan;11(1):710. https://doi.org/10.1038/s41598-020-80605-6 ; PMID: 33436912; PMCID: PMC7804305 , Jan-2021
Articles in Peer-reviewed Journals	Mias GI, Zheng M. "The MathIOmica toolbox: general analysis utilities for dynamic omics datasets." Curr Protoc Bioinformatics. 2020 Mar;69(1):e91. https://doi.org/10.1002/cpbi.91 ; PMID: 31851777 , Mar-2020
Articles in Peer-reviewed Journals	Zheng M, Domanskyi S, Piermarocchi C, Mias GI. "Visibility graph based temporal community detection with applications in biological time series." Sci Rep. 2021 Mar;11(1):5623. https://doi.org/10.1038/s41598-021-84838-x ; PMID: 33707481; PMCID: PMC7952737 , Mar-2021
Articles in Peer-reviewed Journals	Zheng M, Piermarocchi C, Mias GI. "Temporal response characterization across individual multiomics profiles of prediabetic and diabetic subjects." Sci Rep. 2022 Jul 15;12:12098. https://doi.org/10.1038/s41598-022-16326-9 ; PMID: 35840765, PMCID: PMC9284494 , Jul-2022
Articles in Peer-reviewed Journals	Scott RT, Sanders LM, Antonsen EL, Hastings JJA, Park S-m, Mackintosh G, Reynolds RJ, Hoarfrost AL, Sawyer A, Greene CS, Glicksberg BS, Theriot CA, Berrios DC, Miller J, Babdor J, Barker R, Baranzini SE, Beheshti A, Chalk S, Delgado-Aparicio GM, Haendel M, Hamid AA, Heller P, Jamieson D, Jarvis KJ, Kalantari J, Khezeli K, Komarova SV, Komorowski M, Kothiyal P, Mahabal A, Manor U, Garcia Martin H, Mason CE, Matar M, Mias GI, Myers JG, Nelson C, Oribello J, Parsons-Wingerter P, Prabhu RK, Qutub AA, Rask J, Saravia-Butler A, Saria S, Singh NK, Snyder M, Soboczenski F, Soman K, Van Valen D, Venkateswaran K, Warren L, Worthey L, Yang JH, Zitnik M, Costes SV. "Biomonitoring and precision health in deep space supported by artificial intelligence." Nat Mach Intell. 2023 Mar 23;5:196-207. Review. https://doi.org/10.1038/s42256-023-00617-5 , Mar-2023
Articles in Peer-reviewed Journals	Zheng M, Charvat J, Zwart SR, Mehta SK, Crucian BE, Smith SM, He J, Piermarocchi C, Mias GI. "Time-resolved molecular measurements reveal changes in astronauts during spaceflight." Front Physiol. 2023 Jul 14;14:1219221. https://www.frontiersin.org/articles/10.3389/fphys.2023.1219221 ; PMID: 37520819; PMCID: PMC10376710 , Jul-2023
Articles in Peer-reviewed Journals	Xue S, Rogers LRK, Zheng M, He J, Piermarocchi C, Mias GI. "Applying differential network analysis to longitudinal gene expression in response to perturbations." Front Genet. 2022 Oct 17;13:1026487. https://doi.org/10.3389/fgene.2022.1026487 ; PMID: 36324501; PMCID: PMC9618823 , Oct-2022
Articles in Peer-reviewed Journals	Sanders LM, Scott RT, Yang JH, Qutub AA, Garcia Martin H, Berrios DC, Hastings JJA, Rask J, Mackintosh G, Hoarfrost AL, Chalk S, Kalantari J, Khezeli K, Antonsen EL, Babdor J, Barker R, Baranzini SE, Beheshti A, Delgado-Aparicio GM, Glicksberg BS, Greene CS, Haendel M, Hamid AA, Heller P, Jamieson D, Jarvis KJ, Komarova SV, Komorowski M, Kothiyal P, Mahabal A, Manor U, Mason CE, Matar M, Mias GI, Miller J, Myers JG, Nelson C, Oribello J, Park S-m, Parsons-Wingerter P, Prabhu RK, Reynolds RJ, Saravia-Butler A, Saria S, Sawyer A, Singh NK, Snyder M, Soboczenski F, Soman K, Theriot CA, Van Valen D, Venkateswaran K, Warren L, Worthey L, Zitnik M, Costes SV. "Biological research and self-driving labs in deep space supported by artificial intelligence." Nat Mach Intell. 2023 Mar 23;5:208-19. Review. https://doi.org/10.1038/s42256-023-00618-4 , Mar-2023
Articles in Peer-reviewed Journals	Scott RT, Sanders LM, Antonsen EL, Hastings JJA, Park S-m, Mackintosh G, Reynolds RJ, Hoarfrost AL, Sawyer A, Greene CS, Glicksberg BS, Theriot CA, Berrios DC, Miller J, Babdor J, Barker R, Baranzini SE, Beheshti A, Chalk S, Delgado-Aparicio GM, Haendel M, Hamid AA, Heller P, Jamieson D, Jarvis KJ, Kalantari J, Khezeli K, Komarova SV, Komorowski M, Kothiyal P, Mahabal A, Manor U, Garcia Martin H, Mason CE, Matar M, Mias GI, Myers JG, Nelson C, Oribello J, Parsons-Wingerter P, Prabhu RK, Qutub AA, Rask J, Saravia-Butler A, Saria S, Singh NK, Snyder M, Soboczenski F, Soman K, Van Valen D, Venkateswaran K, Warren L, Worthey L, Yang JH, Zitnik M, Costes SV. "Biomonitoring and precision health in deep space supported by artificial intelligence." Nat Mach Intell. 2023 Mar 23;5:196-207. Review. https://doi.org/10.1038/s42256-023-00617-5 , Mar-2023